

A novel temporal–spatial graph neural network for wind power forecasting considering blockage effects

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ABSTRACT

Wind Power Forecasting is crucial for the operational security, stability, and economic efficiency of the power grid, yet it faces significant accuracy challenges due to the variable nature of wind energy and complex interactions within wind farms. This study introduces a novel neural network model specifically designed for Wind Power Forecasting, incorporating both a gated dilated inception network and a graph neural network. This innovative approach enables the concurrent analysis of temporal and spatial features of wind energy, significantly enhancing the forecasting accuracy. A pivotal feature of this model is its unique mechanism to compute the mutual influence between wind turbines, with a particular focus on the blockage effect, a key factor in turbine interactions. The model's efficacy is validated using a real-world dataset, targeting a 48-hour prediction horizon. The experimental outcomes demonstrate that this model achieves superior performance compared to state-of-the-art methods, with a notable improvement of 6.87% in Root Mean Square Error and 8.77% in Mean Absolute Error. This study not only highlights the model's enhanced forecasting capabilities but also emphasizes the importance of integrating spatial and temporal dynamics in wind farms for improving Wind Power Forecasting accuracy.

1. Introduction

Wind energy, recognized for its eco-friendly energy production and wide availability, has become a significant source of renewable energy [1]. The World Wind Energy Association reports that by 2021, the global wind energy capacity reached 837 gigawatts [2], showing a substantial increase, especially in offshore wind capacity, compared to the previous year [3]. However, the variable nature of wind energy poses difficulties in maintaining consistent and reliable power supply to the grid [4]. Therefore, the focus on WPF is growing, as it plays a crucial role in ensuring the power grid's security, stability, and financial viability.

To enhance wind energy production forecasting, physical WPF models have been developed. These models are grounded in detailed mathematical representations of the dynamic interactions within wind turbines and wind farms. The work of Yan et al. [5] exemplifies this approach, demonstrating the intricate mathematical underpinnings required to accurately model these complex systems. The physical models incorporate various parameters and variables that characterize the behavior of wind energy systems under different environmental and operational conditions. However, the application of these physical models comes with its challenges. One of the primary issues, as Hanifi

et al. [6] have emphasized, is the significant computational resources these models demand. The complexity inherent in the mathematical equations that form the basis of these physical models necessitates extensive computational power for effective implementation. This requirement often poses a constraint, particularly in scenarios where rapid and efficient data processing is crucial.

In contrast to physical models, statistical WPF models offer an alternative approach. These models, as illustrated by the research of Liu et al. [7] capitalize on the analysis of extensive historical data sets. They aim to establish predictive correlations between key explanatory variables and wind power generation outcomes. Statistical models typically employ a range of conventional statistical techniques, including ARMA, VAR, and Bayesian methods. These techniques have been particularly adapted for short-term wind power forecasting, leveraging past data to predict future energy outputs. Despite the advantages of statistical models, particularly in terms of their ability to handle large datasets and provide quick forecasts, they are not without limitations. The primary constraint of these models lies in their adaptability and learning capacity. As the prediction horizon extends, the efficacy of these statistical methods tends to diminish. This is due to the inherent limitations of these methods in capturing the complex and

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often non-linear relationships that characterize wind energy production over longer periods. Therefore, while statistical models are valuable for short-term forecasting, their performance in long-term prediction scenarios is an area that requires further research and development.

In the evolving landscape of WPF, deep learning models have recently emerged as a prominent area of study, drawing attention to their advanced representational capabilities. This shift is underscored by the work of researchers like Niu et al. [8] and Liu et al. [9] who have explored the potential of deep learning in this domain. These models diverge from traditional approaches by focusing on developing mapping functions that directly relate input variables to wind power generation outputs, instead of depending on pre-established physical assumptions. The core advantage of deep learning models in WPF lies in their ability to capture complex, non-linear relationships within data. Unlike conventional models that may rely heavily on linear assumptions or simplified representations of the wind energy process, deep learning models can learn from the data itself, identifying intricate patterns and relationships that might otherwise be overlooked. This capability makes them particularly suited to handling the unpredictability and complexity inherent in wind power generation. However, the journey towards fully harnessing the potential of deep learning in WPF is not without its challenges. A significant hurdle lies in addressing the high levels of uncertainty and non-linearity associated with wind power. These aspects introduce complexity in modeling and forecasting, which deep learning models must effectively navigate. While deep learning offers a powerful tool for representation and prediction, the development of these models often requires careful consideration and integration of inductive biases that guide the learning process. Inductive biases help the model to make reasonable assumptions or generalizations from the data, which is especially important in scenarios where data may be limited or highly variable.

Wind power prediction models each have their unique strengths and weaknesses. In an effort to maximize their benefits and minimize their limitations, researchers like Ma et al. [10] have developed hybrid models that combine various model features, showing promising outcomes. Despite these advancements, it is crucial to consider the changing environmental conditions around wind farms. Wind behavior tends to follow specific patterns over time and space, suggesting the importance of using historical data in a sequential manner for better prediction accuracy. Recent studies, including those by Zhang et al. [11] have started to incorporate spatial statistics related to the environment around wind turbines. Some have even used data from nearby turbines or towers to enhance predictions. However, these methods typically focus on individual turbines and do not fully exploit the spatial characteristics of the entire wind farm, as pointed out by Fan et al. [12] This limitation becomes more pronounced in wind farms where turbines are spread over a large area. Therefore, identifying features that effectively capture both the temporal and spatial dynamics of wind farms is essential for improving the accuracy of wind power predictions. Such an approach would ensure a more holistic understanding of wind patterns, leading to more reliable and efficient energy forecasting.

GCN has become increasingly popular for predicting wind turbine power generation, particularly because they are adept at capturing both temporal and spatial features simultaneously. Traditional methods in this area often use a distance-based approach to estimate how wind turbines influence each other. However, recent research highlights that there are two primary types of interactions between wind turbines: blockage and wake effects. These interactions are intricately connected to the turbines' relative positions and the surrounding environmental conditions, and they can significantly affect power generation.

The blockage effect occurs when a turbine obstructs the wind flow to another, while the wake effect refers to the reduced wind speed and increased turbulence behind a turbine. Both of these effects can substantially decrease the efficiency and power output of wind turbines. Therefore, to enhance the accuracy of wind turbine power generation

Table 1

Table of abbreviation.

No.	Acronym	Meaning
1	WPF	wind power forecasting
2	RMSE	Root Mean Square Error
3	MAE	Mean Absolute Error
4	ARMA	AutoRegressive Moving Average
5	VAR	Vector Autoregression
6	GNN	Graph convolutional networks
7	NWP	Numerical Weather Prediction
8	GRAPES	Global Regional Assimilation and prediction Enhances System
9	ARIMA	Autoregressive Integrated Moving Average Model
10	SVM	Support Vector Machines
11	RSVM	Robust Support Vector Machines
12	DBN	Deep Belief Network
13	CNN	Convolutional Neural Network
14	ICA	Independent Component Analysis
15	RNN	Recurrent Neural Network
16	CFD	Computational Fluid Dynamics
17	DTW	Dynamic Time Warping
18	SCADA	Supervisory Control and Data Acquisition

predictions, it is crucial to integrate a more precise representation of these effects into models. By doing so, models can more accurately reflect the complex dynamics of wind flow and turbine interactions, leading to better prediction outcomes. This approach marks a significant advancement in the field, moving beyond simplistic distance-based estimations to a more nuanced understanding of turbine interactions.

In this work, a novel temporal-spatial graph neural network is proposed for WPF considering the blockage and wake effects. The main contributions of this work are as follows:

- A novel temporal-spatial graph neural network for wind energy forecasting considering blockage effects is proposed. This network uniquely combines a gated dilated inception layer with a graph neural network, allowing it to learn both time-related and space-related features at the same time.
- To study the interactions between wind turbines, a new method for the calculation of the adjacency matrix is proposed. The performance of the model is demonstrated by using the actual data from a wind farm. Compared to the state-of-the-art models, the proposed model shows its effectiveness in forecasting wind energy.

2. Related works

2.1. Physical models in WPF

Physical forecasting methods, particularly NWP [13,14], are essential in wind power forecasting for their detailed simulation of weather dynamics. NWP models solve complex hydrodynamic and thermodynamic equations, providing an extensive numerical analysis based on actual atmospheric conditions [15]. Key parameters such as temperature, air density, pressure, and geographical features are integrated to formulate accurate wind speed predictions [7].

Notable NWP models like the GRAPES [16], as shown in Table 1, integrated forecasting system, and Weather Research and Forecasting Model have been developed to cater to varied meteorological needs across different regions, each tailored to their specific environmental conditions [17]. However, the complexity and data intensity of these models demand significant computational resources, which can limit their applicability, particularly in short-term forecasting scenarios [18].

2.2. Statistical techniques in WPF

Statistical methods play a crucial role in short-term forecasting within wind power, especially in sectors dealing with large, interconnected datasets [19]. There are two main methods: time series approaches and Kalman filter techniques [20].

Time series approaches use historical wind speed data, employing models such as AR, ARMA, ARIMA, and Bayesian methods [21]. Each model varies in complexity, suited to different forecasting needs. Innovations in this area include the integration of ICA with AR models for improved forecast precision [22] and the use of ARMA for short-term wind power prediction with reduced error margins [23]. However, these methods necessitate consistent wind data and may struggle with significant data fluctuations.

The Kalman filter, an advanced autoregressive algorithm based on historical data, has been effective in short-term wind speed prediction since its introduction in 1985 [24]. It enhances forecast accuracy by incorporating state and measurement equations. Despite its effectiveness, the Kalman filter's performance is contingent on the precision of these equations, particularly in predicting variable wind speeds.

2.3. Machine learning innovations in WPF

2.3.1. Conventional machine learning models

Machine learning has emerged as a dominant force in wind speed prediction, thanks to its robust data analysis and pattern recognition capabilities. SVM and RSVM are key techniques in this domain, known for their effectiveness in processing real-time data from wind power plants [15]. Hybrid models merging machine learning with statistical methods have also shown significant improvements in forecast accuracy [25].

2.3.2. Neural network-based deep learning models

The advent of deep learning, powered by advancements in neural networks and computing technology, has brought about a new era in wind speed prediction [26]. Deep learning models, such as DBM, CNNs, and hybrid networks, have shown exceptional performance in both short-term and long-term forecasting, outperforming traditional methods [27].

Recent research has focused on developing methods that analyze spatial and temporal data to predict wind farm performance [28]. Challenges arise, especially when dealing with extensive, widely distributed wind farms, as the complexity of the networks can impede the effectiveness of models like graph-deep neural networks. Alternative approaches, like joint distribution models using copula theory and hybrid neural networks, have been explored, but they face limitations in processing capacity and static correlation analysis [15,29,30].

According to the literature, the deep neural networks show better performance than the physical-based models, statistical models, and machine learning models. These deep neural network models, especially graph neural networks, highlight the necessity to consider the physical locations of wind farms and manage the extensive data from multiple sources. The dynamic nature of spatio-temporal correlations remains a critical challenge to address in wind power forecasting.

3. Method

3.1. Problem statement

The primary objective of this study is to formulate an advanced predictive model that specializes in forecasting wind energy production, utilizing complex spatiotemporal data. This research utilizes a comprehensive dataset comprised of sequential time series data, represented as pairs $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, each pair being uniquely linked with its corresponding geographical coordinates $\{loc_1, loc_2, \dots, loc_n\}$. Within

this framework, x_i embodies a multidimensional sequence that encapsulates temporal attributes, y_i represents the observed wind energy output at the moment x_i , and loc_i indicates the precise geographical positioning of the relevant wind turbines at that given time. The foremost aim of this endeavor is to construct robust predictions for wind energy generation over forthcoming h time intervals, leveraging the intricacies and comprehensive nature of the dataset at hand.

The problem under investigation is formally articulated as follows:

$$\hat{y}_{n+1}, \hat{y}_{n+2}, \dots, \hat{y}_{n+h} = f(x_{n+1}, x_{n+2}, \dots, x_{n+h}, loc_{n+1}, loc_{n+2}, \dots, loc_{n+h}) \quad (1)$$

In this equation, $\hat{y}_{n+1}, \hat{y}_{n+2}, \dots, \hat{y}_{n+h}$ symbolize the forecasted outputs for wind power generation across the ensuing h time intervals. The function f encapsulates the algorithm of the predictive model, tasked with the computation of these estimations.

The intrinsic complexity of this problem emanates from the dynamic interaction between the spatiotemporal elements inherent in the dataset. The precision of the forecasts hinges on the model's capacity to adeptly synthesize both the temporal sequences denoted by x_i and the geographic context conveyed by loc_i . Mastery in assimilating and leveraging these detailed complexities is crucial in the creation of an effective, accurate, and dependable prediction framework for wind power generation.

3.2. Overview

The architecture of the proposed model, illustrated in Fig. 1, is organized into three interconnected modules: data preprocessing, temporal-spatial neural network, and energy prediction. Initially, the data preprocessing module undertakes the crucial task of refining and organizing the dataset, ensuring it is optimized for processing by the subsequent stages of the model. Following this, the temporal-spatial neural network module, which is bifurcated into two distinct sub-modules – the temporal neural network and the graph neural network – comes into play. The temporal neural network is dedicated to distilling temporal patterns from historical datasets, whereas the graph neural network focuses on dissecting spatial information and evaluating the impact of obstructive effects. Conclusively, the energy prediction module embarks on the pivotal role of forecasting wind power generation, encompassing both single-step and multi-step predictive analyses. This structured approach integrates sophisticated neural network technologies to unravel the complexities of wind energy prediction.

3.3. Data preprocessing

The data preprocessing stage encompasses a series of meticulous procedures to ensure data integrity and reliability:

- **Zero Values:** This procedure involves rectifying the dataset by setting any active power or reactive power values that are less than zero to zero. This adjustment is crucial for maintaining data consistency and accuracy.
- **Missing Values:** Due to unforeseen circumstances, certain data points may not be captured by the SCADA system. To address this, the K-nearest neighbor imputation technique is employed to fill in these missing values. However, it is important to note that these imputed values are excluded from the model evaluation process to maintain the integrity of the evaluation.
- **Unknown Values:** There are instances where the actual power generation by wind turbines remains unknown, particularly during periods of maintenance or scheduled downtime. To identify these unknown values, two specific conditions are established: Condition 1: If the active power is less than or equal to 0 and the wind speed exceeds 2.5 m/s. Condition 2: If any pitch angle exceeds 89 degrees.

Data points meeting these conditions are classified as 'unknown' and are not considered in the model evaluation to ensure accuracy.

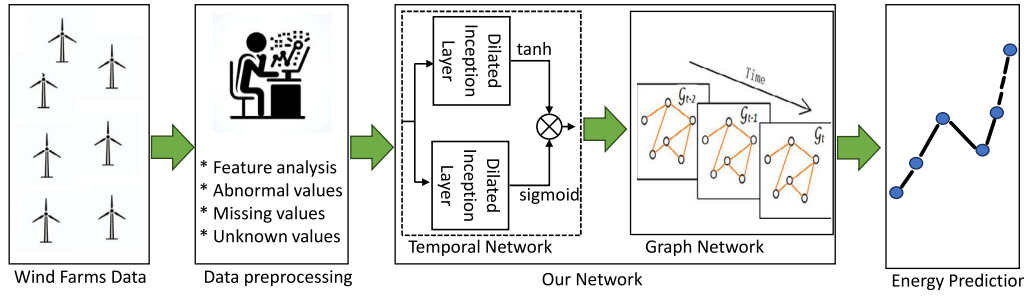


Fig. 1. Overview of our proposed method.

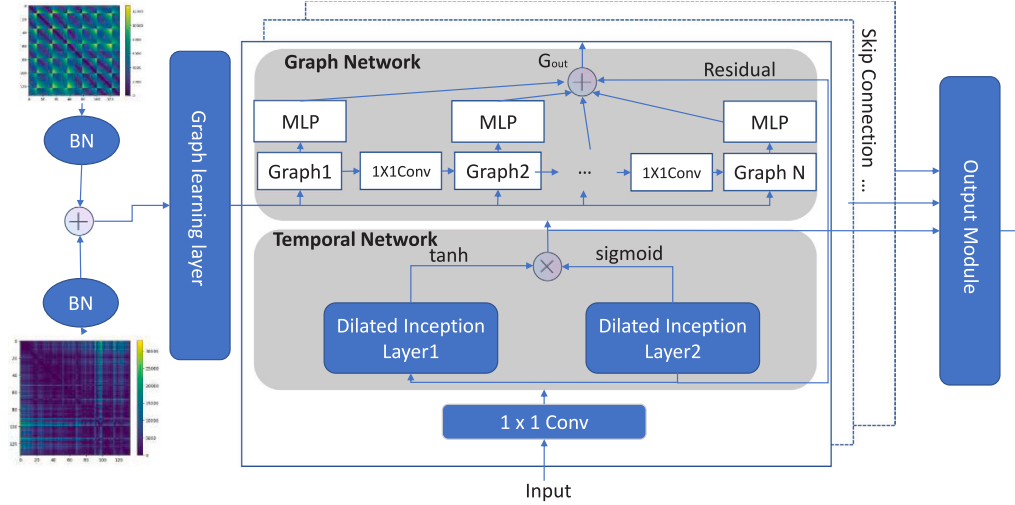


Fig. 2. Detailed information of our proposed model.

- **Abnormal Values:** The dataset may contain abnormal values that can skew the model's performance. To mitigate this, two criteria are set to identify and exclude such values from the model evaluation: Rule 1: Nacelle direction falling outside the range of $[-720, 720]$ degrees. Rule 2: The angle between the wind direction and turbine nacelle position lying outside the range of $[-180, 180]$ degrees.

Each of these steps plays a pivotal role in refining the dataset, thereby laying a robust foundation for the subsequent stages of the temporal-spatial neural network and energy prediction modules.

3.4. Temporal neural network

The forecasting of wind power generation is notably challenging due to the dynamic and unpredictable nature of the power grid system, as highlighted by Wang et al. [31]. This system is marked by its complex characteristics, including abrupt changes and unpredictable patterns in wind power generation, primarily attributed to its inherent randomness. To effectively navigate this complexity, we introduce a temporal network adept at adapting to the variable patterns and sudden shifts in wind power, which may arise from external influences such as wind turbine renovations or adjustments in active power control. This network is designed to discern and learn from prevailing representative patterns, thereby enhancing its forecasting capabilities.

Traditional RNN methodologies, while proficient in handling sequential data, often encounter limitations like vanishing gradients and substantial computational demands due to their recursive structure. To circumvent these challenges, we adopt the stacked dilated causal convolution layers, a concept pioneered by Yu et al. [32]. This approach utilizes dilated convolutional layers, enabling our temporal network to

cover extensive temporal ranges without succumbing to the vanishing gradient issue commonly associated with RNN-based models.

In our model, the input is denoted as $X \in \mathbb{R}^{F \times N \times W}$, where F indicates the feature dimension, N represents the number of nodes (wind turbines), and W signifies the window size of historical data. As shown in Fig. 2, the model first computes the initial temporal feature through a convolutional layer. Following this, the Dilated Inception layer, as described by Yu et al. is employed. This layer is instrumental in capturing long-term temporal dependencies and employs a gating mechanism to achieve this. The operation of the Dilated Inception layer is encapsulated by the equation:

$$x_o = \tanh(x_i * \alpha) + \delta(x_i * \beta) \quad (2)$$

In this equation, x_i is the input to the Dilated Inception layer, and x_o is the output. Notably, this formulation omits the calculation bias for simplicity. The Dilated Inception module comprises K parallel layers, each linked through residual connections, enhancing the model's capability to capture and process complex temporal patterns inherent in wind power generation data.

3.5. Graph neural network

3.5.1. Graph construction considering the blockage effects

The learning of spatial information among wind turbines, a critical aspect of this study, is accomplished using a graph network. The initial hurdle in deploying graph networks for spatial feature learning involves the formulation of an adjacency matrix. Conventionally, the creation of an adjacency matrix predominantly relies on assessing the relative positional relationships among various wind turbines. However, this traditional method falls short as it does not take into account the

blockage effect, which can significantly influence the efficiency and output of wind turbines.

The blockage effect is a crucial factor in understanding the interactions between wind turbines. This effect occurs when the airflow to a turbine is obstructed by neighboring turbines, leading to a reduction in wind speed and consequently, the energy output of the affected turbine. Ignoring the blockage effect can lead to inaccurate assessments of turbine efficiency and spatial interactions.

To address this, our approach involves a more nuanced method of calculating the adjacency matrix that extends beyond mere positional relationships. This method incorporates the blockage effect, thereby providing a more comprehensive and realistic representation of the spatial relationships and interactions among wind turbines. By integrating the blockage effect into the adjacency matrix calculation, the graph network is better equipped to accurately learn and represent the spatial dynamics within a wind farm. This improved understanding is essential for enhancing the accuracy and effectiveness of wind power generation forecasting models.

CFD simulations are a prevalent tool for assessing the blockage effects in wind turbines, as noted by Branlard et al. [33]. Despite their usefulness, CFD simulations have limitations in accurately measuring the blockage effects of wind turbines. Additionally, while the wake effect among wind turbines is significant, it cannot be directly quantified from the available data. However, the interactions between wind turbines, whether among adjacent turbines or groups of turbines, are implicitly present in the historical operational data of these turbines.

To overcome these challenges, our study introduces an innovative methodology for constructing the adjacency matrix. This process involves two main stages:

1. Creating a Geographic Adjacency Matrix: In this stage, the geographical information of each wind turbine is utilized to construct a matrix. This approach ensures that the geographic location of each turbine is fully integrated into the matrix, providing a detailed spatial perspective.

2. Formulating a DTW-Based Graph: To model the interaction effects between wind turbines, we create a graph using the historical operational data of the turbines. This graph is based on the DTW algorithm, as described by Yi et al. [34]. DTW is a dynamic programming algorithm that calculates the minimum distance between two time series, making it ideal for understanding the temporal relationships in turbine data.

By employing the DTW-based graph, it becomes possible to train a graph neural network that can discern the mutual influence effects between wind turbines. This network, using the DTW-based graph as input, learns to perform graph convolutions. These convolutions are key in aggregating information from neighboring nodes, thereby enabling the network to detect and understand complex interactions between turbines. Such interactions, including blockage and wake effects, are not directly observable in the data but significantly affect turbine performance. Through this method, the model can effectively capture these intricate dynamics, providing a more comprehensive and accurate understanding of wind turbine interactions for enhanced forecasting of wind power generation.

3.5.2. Graph learning

In this study, the graph convolutional module incorporates a sophisticated ‘mix-hop’ propagation layer, which operates through two principal learning phases within the context of the established correlation graph, as illustrated in Fig. 2. These phases are crucial in refining and utilizing the spatial information from the graph network effectively.

1. Information Propagation Step: The first phase entails the propagation of information, mathematically expressed as follows:

$$H^{(k)} = \alpha H_{in} + (1 - \alpha) \hat{A} H^{(k-1)} \quad (3)$$

Here, H_{in} is the initial output derived from the temporal neural network, and $H^{(k)}$ signifies the output after the k th propagation iteration. The coefficient α plays a pivotal role, dictating the balance between retaining input features and integrating the graph’s features. The matrix $\hat{A}$ represents the normalized Laplace matrix of the correlation graph, calculated as $D^{-1}(A + I)$, where $D_{ij} = 1 + \sum_i A_{i,j}$ and I is the identity matrix introducing self-connections to maintain the original feature integrity. The graph A is constructed as detailed in the preceding sections.

2. Information Selection Step: The second phase involves an adaptive selection process, focusing on the outcomes of the K -step information propagation:

$$H_O = \sum_{i=0}^K H^{(i)} W^{(i)} \quad (4)$$

In this equation, K represents the total number of propagation steps, and $W^{(k)}$ are the learnable parameters specific to each step. The final output, denoted as H_O , encapsulates the hidden states of the mix-hop propagation graph convolution module.

These two steps in the mix-hop propagation layer are integral to the model’s capability to effectively process and interpret the complex spatial relationships inherent in the wind turbine network. The combination of information propagation and selection ensures that the graph convolutional module captures both immediate and broader spatial correlations, leading to a more nuanced understanding and accurate prediction of wind power generation.

3.6. Energy prediction

In this research, the proposed model is meticulously crafted and trained to forecast wind power generation with a lead time of two days. This forecasting is grounded on historical data spanning a continuous sequence of 288 data points, symbolized as $W = 288$. This duration effectively represents a 48-hour window, during which the model leverages past data to project wind power output for the forthcoming 48 h. The versatility of the model is highlighted by its capacity to adjust the window size W to meet specific requirements.

To operationalize this design, the model incorporates a convolutional layer that plays a key role in processing the input data. In the training phase, the model employs the Huber loss function, as elucidated by Huber [35]. This loss function is particularly chosen for its reduced sensitivity to outliers compared to the conventional squared error loss function. The mathematical expression of the Huber loss function is as follows:

$$L_{\delta}(y, f(x)) = \begin{cases} \frac{1}{2}(y - f(x))^2, & \text{if } |y - f(x)| \leq \delta \\ \delta|y - f(x)| - \frac{1}{2}\delta^2, & \text{otherwise} \end{cases} \quad (5)$$

In this equation, y denotes the actual value, $f(x)$ represents the predicted value, and δ is a hyperparameter that dictates the threshold for transitioning the loss function from a quadratic form to a linear form. When the absolute difference between the actual and predicted values is within the limit set by δ , the quadratic term is utilized for loss calculation. Conversely, if this difference exceeds δ , the loss is computed using the linear term, effectively diminishing the impact of outliers. This property renders the Huber loss function more resilient to outliers than the squared error loss, thereby enhancing the robustness and accuracy of the model in predicting wind power generation.

4. Experiment

In our study, we evaluated a wind power forecasting model on a server equipped with an NVIDIA V100 GPU, focusing on predicting future wind power supply based on past data with sets of 288 and 144 data points. The model training used the Adam optimizer with a learning rate of 0.001, across 30 epochs, and a batch size of 64. For stability, we implemented gradient clipping with a threshold of 5 and

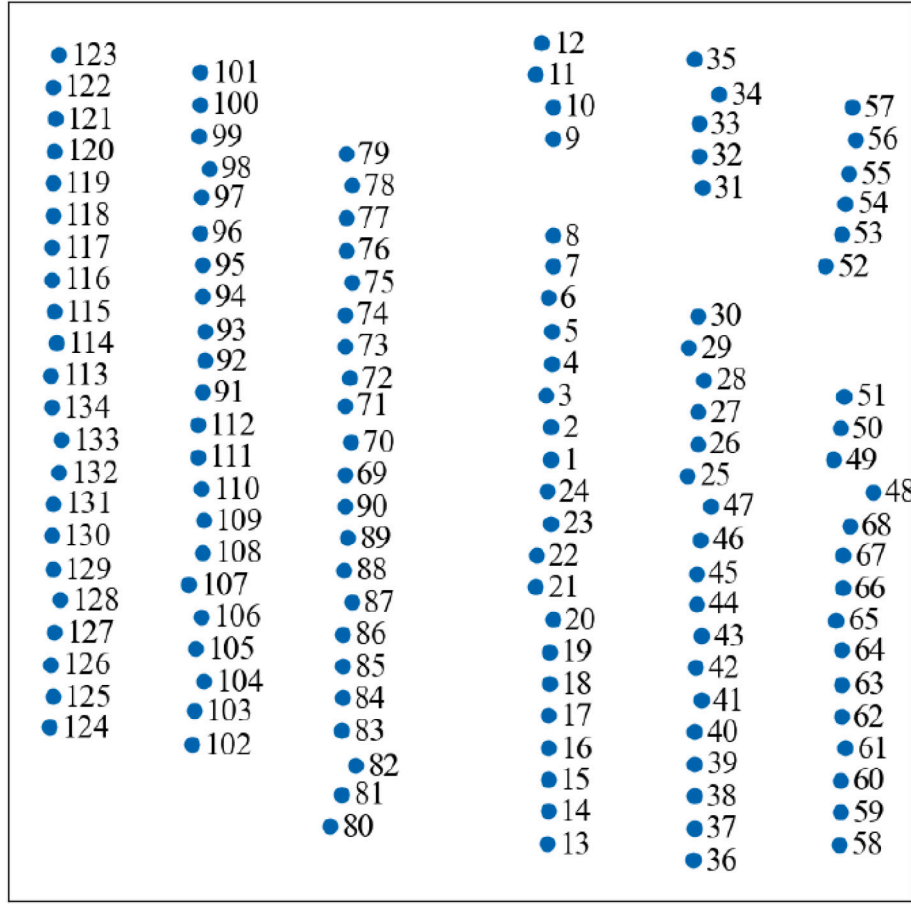


Fig. 3. Spatial distribution of all wind turbines (x and y are with the meter unit).

Table 2

Dataset specification.

No	Variable name	Physical meaning
1	TurbID	Wind turbine ID
2	Day	Day of the record
3	Tmstamp	Created time of the record
4	Wspd (m/s)	The wind speed recorded by the anemometer
5	Wdir (°)	The angle between the wind direction and the position of turbine nacelle
6	Etmp (°C)	Temperature of the surrounding environment
7	Itmp (°C)	Temperature inside the turbine nacelle
8	Ndir (°)	Nacelle direction, i.e., the yaw angle of the nacelle
9	Pab1 (°)	Pitch angle of blade 1
10	Pab2 (°)	Pitch angle of blade 2
11	Pab3 (°)	Pitch angle of blade 3
12	Prtv (kW)	Reactive power
13	Patv (kW)	Active power (target variable)

employed the Huber loss function with a delta value of 5 to lessen the impact of outliers. The model's architecture combined three gated temporal convolutional modules with spatial graph convolutional modules, configured with hidden and skip connection dimensions of 32 and 64, respectively. The graph convolutional module, crucial for processing spatial data, operates with 2 steps. This comprehensive approach ensured precise and efficient forecasting in wind power generation.

4.1. Dataset

The dataset employed in this research was sourced from the Baidu KDD CUP 2022 [36]. It comprises SCADA data, systematically collected at ten-minute intervals from each turbine within a 134-unit wind farm, culminating in a comprehensive compilation of 4,727,520 individual data records over a span of 245 days. This extensive dataset encompasses a range of both external and internal variables, each playing a pivotal role in the accurate depiction of the operational status of the wind turbines. Moreover, the dataset includes a detailed spatial representation of the relative positioning of the turbines within the wind farm, as depicted in Fig. 3. Furthermore, a thorough delineation of the dataset's specifications is presented in Table 2.

4.2. Evaluation metrics

In this study, the RMSE and MAE serve as the principal evaluation metrics. For an individual wind turbine, denoted as i , at a specific time point t_k , the RMSE and MAE are calculated by juxtaposing the forecasted power output $\hat{Y}_{t_k+t}$ against the actual power output Y_{t_k+t} across a future interval encompassing 288 time steps ($t \in [1, 288]$).

The formulas for RMSE and MAE are expressed as follows:

$$RMSE_{t_k}^i = \sqrt{\frac{\sum_{t=1}^{288} (Y_{t_k+t} - \hat{Y}_{t_k+t})^2}{288}} \quad (6)$$

$$MAE_{t_k}^i = \frac{\sum_{t=1}^{288} |Y_{t_k+t} - \hat{Y}_{t_k+t}|}{288} \quad (7)$$

It is important to note that during the computation of RMSE and MAE, any data points that are missing, unknown, or deemed abnormal within the original dataset are systematically excluded.

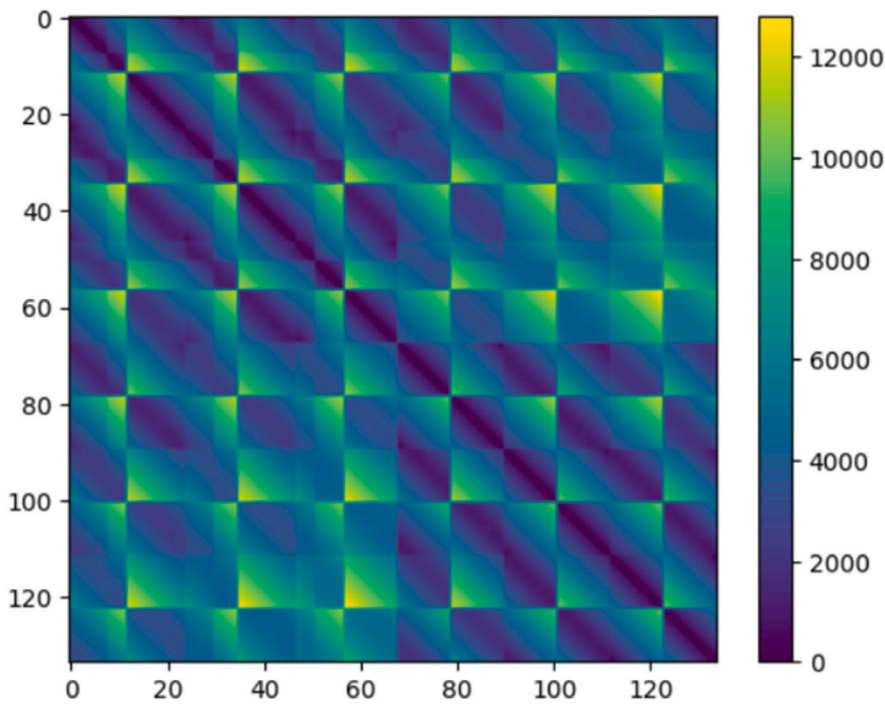


Fig. 4. Geographic interactions between wind turbines.

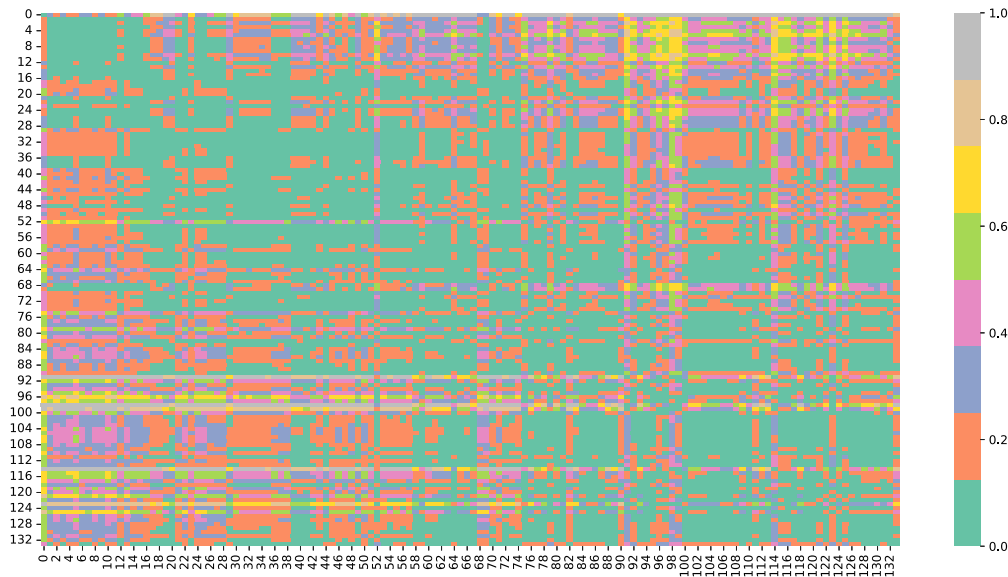


Fig. 5. The graph considering the interaction effects between wind turbines.

The aggregate score for this evaluation is derived by averaging the RMSE and MAE across all wind turbines over a series of K designated timestamps T , which is articulated as:

$$score = \frac{1}{2K \times 134} \sum_{k=0}^K \sum_{i=1}^{134} (RMSE_{T(k)}^i + MAE_{T(k)}^i) \quad (8)$$

4.3. Graph construction

Fig. 4 illustrates the adjacency matrix representing the relative spatial positioning of each wind turbine within the dataset under examination. This visualization distinctly highlights the varying influence levels of the wind turbines on each other based on their proximity. It is evident from the matrix that turbines in closer adjacency exert a more

substantial effect on each other, while the impact diminishes as the spatial separation increases.

Furthermore, the analysis of Fig. 5 reveals intriguing insights into the hidden patterns within the historical data of the wind turbines. This analysis uncovers that certain wind turbines, although geographically distant, demonstrate remarkably similar power generation patterns. Conversely, some turbines that are in close spatial proximity exhibit less congruent behavior patterns. For instance, turbines 2, 3, 4, 5, 6, and 7, which almost align along the same y -axis, do not show a high degree of behavioral similarity. In stark contrast, turbines 1, 53, 60, 80, and 92, despite their widespread spatial distribution, display a strikingly similar operational behavior. This phenomenon is hypothesized to be influenced by the directional impact of wind on different turbines.

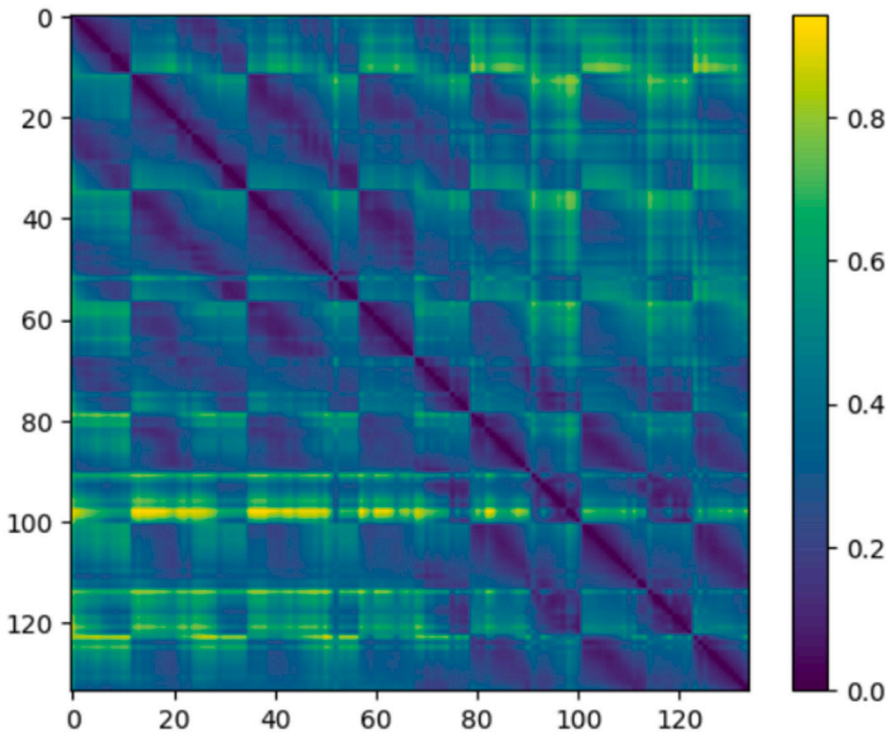


Fig. 6. The normalized fusion graph using the above graphs.

In an effort to synergize the insights gleaned from both the aforementioned graphical representations, a novel fusion graph has been constructed, as depicted in Fig. 6. This innovative graph, serving as an input for the graph neural network, harnesses the spatial interconnections between the wind turbines to enhance the precision of wind power production forecasts. The graph neural network is adept at propagating information across the network, thereby capturing the intricate dependencies among the turbines. This capability is then integrated into the predictive model. By doing so, the model is able to more effectively comprehend the complex interrelationships among the wind turbines, thereby significantly elevating the accuracy of the wind power generation predictions.

4.4. Comparison with state-of-the-art methods

- **Historical average:** This technique relies on the average of historical data from the training set for each wind turbine to forecast values in the test dataset.
- **Moving average:** This method utilizes a moving average strategy, applying a window size of 288 (corresponding to 48 h), to predict the output of each wind turbine.
- **GRU:** A well-established deep learning approach for time series analysis, the GRU, a variant of the RNN, employs gating mechanisms to more effectively capture long-term dependencies within time series data [37].
- **GNN:** This approach leverages a graph neural network to discern and utilize the concealed spatial relationships among the wind turbines in the farm [36].
- **LightGBM:** A gradient boosting framework utilizing tree-based learning algorithms, LightGBM has demonstrated exceptional performance in the realm of time series forecasting [38].
- **MDLinear (single model):** This straightforward yet potent method uses a multi-dense linear layer architecture to model time series data. It has been noted for its efficacy, often surpassing many transformer-based models in similar tasks [39].

Table 3

Comparison results with State-of-the-art methods.

Method	RMSE	MAE	Score
Historical average	56.72	47.86	52.29
Moving average	61.56	50.62	56.09
GRU	55.13	45.77	50.45
GNN	55.39	47.15	51.27
LightGBM	53.05	44.89	48.97
MDLinear (single model)	56.74	48.32	52.53
Modified DLinear	53.40	45.53	49.46
XTGN	54.54	46.50	50.52
Fused model	53.74	45.86	49.80
Ours	49.64	41.27	45.46

- **Modified DLinear:** An adaptation of the MDLinear method, this variant incorporates an extra linear layer to more comprehensively integrate information from all input features [40].
- **XTGN (extreme Temporal Gated Network):** A novel model, XTGN blends gated temporal convolutional modules with spatial graph convolutional modules, thereby capturing both the temporal and spatial dependencies inherent in time series data [40].
- **Fused model:** This hybrid model merges the capabilities of the Modified DLinear and XTGN models, aiming to enhance the accuracy and reliability of wind power forecasting [40].

Table 3 in the study provides a detailed comparative analysis of various forecasting methodologies, juxtaposing the proposed approach with other state-of-the-art techniques in the field.

The outcomes of this comparative study unequivocally demonstrate the superior performance of the proposed method across all three evaluation metrics: RMSE, MAE, and the overall score. This method achieves the lowest values in each of these metrics, underscoring its effectiveness in wind energy prediction.

Among the competing methods, LightGBM emerges as the second most effective approach, showcasing commendable performance. It is closely followed by the Modified DLinear and the Fused model, both of

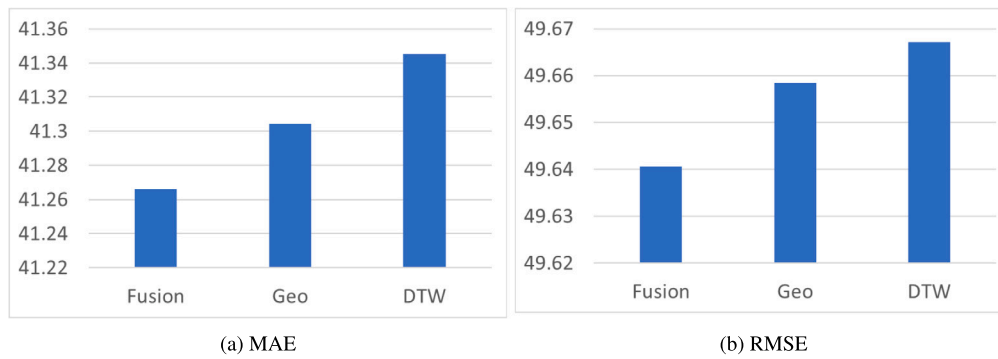


Fig. 7. Ablation study.

which also display significant forecasting capabilities. On the other end of the spectrum, the historical average and moving average methods exhibit the least favorable results. This outcome highlights the necessity of employing more sophisticated and advanced modeling techniques for the accurate prediction of wind energy.

In summary, the results underscore the effectiveness of the proposed method, particularly its use of a graph neural network. This approach enables it to discern and leverage the hidden interconnections among wind turbines, thereby enhancing the accuracy and reliability of wind energy forecasts.

4.5. Ablation study

The significance of the proposed graph fusion technique in this research is demonstrated through the comparison of three distinct variants:

1. **Fusion:** This is the proposed method, which integrates the elements of both geographic positioning and semantic relationships in its analysis.
2. **Geo:** In this variant, only the geographic position of the wind turbines is considered for the analysis.
3. **DTW:** This method solely utilizes the semantic relationship between the data points, without taking geographic positioning into account.

The comparative results (see Fig. 7) clearly indicate that the Fusion method, which combines both geographic and semantic data, significantly outperforms the other two variants in terms of MAE and RMSE. This superior performance underscores the effectiveness of integrating both spatial and semantic aspects in wind power forecasting.

Conversely, the DTW method, which relies exclusively on semantic relationships, exhibits the least favorable performance among the three. This outcome suggests that while semantic relationships are important, they are insufficient on their own for achieving optimal forecasting accuracy. The inclusion of geographic data appears to be a critical factor in enhancing the predictive capability of the model.

Overall, these findings emphasize the importance of the graph fusion approach proposed in this study. By effectively synthesizing geographic and semantic information, the Fusion method demonstrates a more comprehensive and accurate approach to wind power forecasting, highlighting its potential utility in improving the efficiency and reliability of renewable energy systems.

4.6. Sensitivity analysis

The sensitivity analysis conducted in this study aims to ascertain the impact of various hyperparameters on the performance of the model. The findings, illustrated in Fig. 8, provide valuable insights into how these hyperparameters influence the model's forecasting accuracy.

1. **Convolution Channels (conv_channels):** When the number of convolution channels is set to 32, the model achieves its optimal performance, evidenced by the lowest MAE and RMSE values, recorded at

41.2661 and 49.6405 respectively. This setting denotes the peak accuracy of the model's predictions. Notably, as the number of conv_channels is varied, there is an observed trend where the MAE and RMSE values decrease initially, followed by an increase. This trend suggests that there is an optimal point beyond which increasing the number of conv_channels does not contribute to, and may even detract from, the model's accuracy.

2. **Residual Channels (residual_channels):** A similar trend is observed with the number of residual channels. The model's highest accuracy is again achieved when the residual_channels are set to 32, with the MAE and RMSE values mirroring those obtained with the optimal conv_channels setting (41.2661 and 49.6405, respectively). This consistency in optimal performance across different hyperparameters reaffirms the robustness of the model under this particular parameter configuration. As the number of residual_channels increases, there is a general upward trend in both MAE and RMSE values, indicating a decrease in prediction accuracy.

3. **Skip Channels (skip_channels):** Setting the number of skip channels to 64 results in the lowest MAE and RMSE values (41.2661 and 49.6405, respectively), signifying the most accurate predictions under this setting. However, as with the previous parameters, increasing the number of skip_channels beyond this point generally leads to higher MAE and RMSE values, indicating a reduction in the predictive accuracy of the model.

These observations highlight the critical role of hyperparameter tuning in achieving optimal performance in wind power forecasting models. The specific settings of conv_channels, residual_channels, and skip_channels play a significant role in the model's ability to accurately forecast wind power generation, underscoring the importance of meticulous hyperparameter optimization in developing effective predictive models.

5. Conclusions

Wind power, as a burgeoning source of renewable energy, has garnered global attention, leading to WPF becoming a pivotal research area. This focus is essential for enhancing the security, stability, and economic viability of the power grid. In recent times, graph convolutional networks have gained prominence for their ability to predict wind turbine power generation, uniquely capturing both temporal and spatial features. However, a significant limitation of many current methodologies is their reliance on distance-based approaches, which may not fully encapsulate the complex interactions between turbines, such as blockage and wake effects. Accurately representing these influences is crucial for refining the precision of wind turbine power generation forecasts. Addressing this gap, this study introduces an innovative neural network model for WPF, integrating a gated dilated inception framework with a graph neural network. This model is adept at concurrently learning temporal and spatial features. A distinctive feature of this work is the development of a novel mechanism to calculate

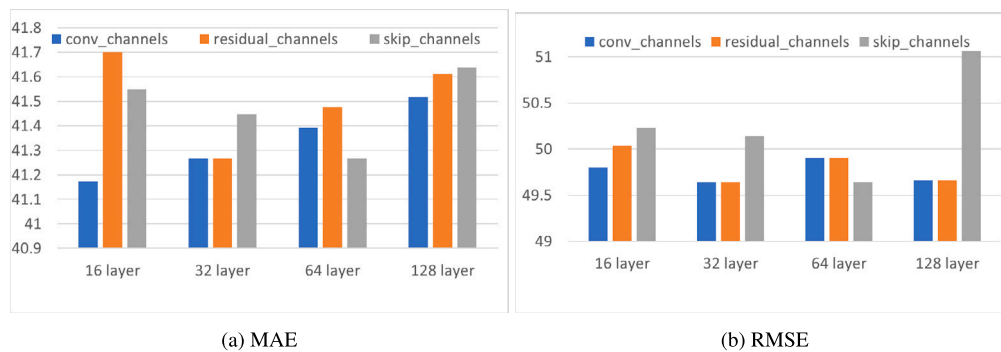


Fig. 8. Sensitivity analysis.

the inter-turbine influence, which adeptly captures spatial characteristics and enhances forecasting accuracy by considering the specific wind effects on each turbine. The efficacy of the proposed model is substantiated through validation on a real-world dataset for wind power forecasting. The experimental results compellingly demonstrate that this model surpasses existing state-of-the-art methods in forecasting accuracy. More specifically, our model has a notable improvement of 6.87% in Root Mean Square Error and 8.77% in Mean Absolute Error. Overall, this research marks a significant stride in advancing the accuracy and reliability of WPF models, a critical step forward in promoting the sustainable evolution of the power grid infrastructure.

The future work will focus on designing the lightweight version of the forecasting model, as there are not enough computational resources in wind farms. More importantly, there is a need for improving the multiscale features from the sensor data of wind turbines.

CRedit authorship contribution statement

Hong Qiu: Writing – original draft, Writing – review & editing. **Kaikai Shi:** Data curation, Formal analysis. **Renfang Wang:** Funding acquisition, Investigation. **Liang Zhang:** Conceptualization, Methodology, Visualization. **Xiufeng Liu:** Methodology, Project administration, Resources. **Xu Cheng:** Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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